

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

The Core Problem

Existing approaches blindly inject gold trajectories, external feedback, and domain knowledge without considering situational needs.

This 'flood irrigation' leads to:

- Fragile planning on unexpected inputs
- Pattern collapse under distribution shift
- Unnecessarily high inference costs

Humans possess situational self-awareness:

We naturally assess whether we can handle a situation directly, need to pause and reflect, or must seek external help. KnowSelf brings this to LLM agents.

Trajectory Imitation

ReAct

Copies gold action sequences regardless of situation

Trial-and-Error

Reflexion

Applies reflection uniformly even when unnecessary

Knowledge-Augmented

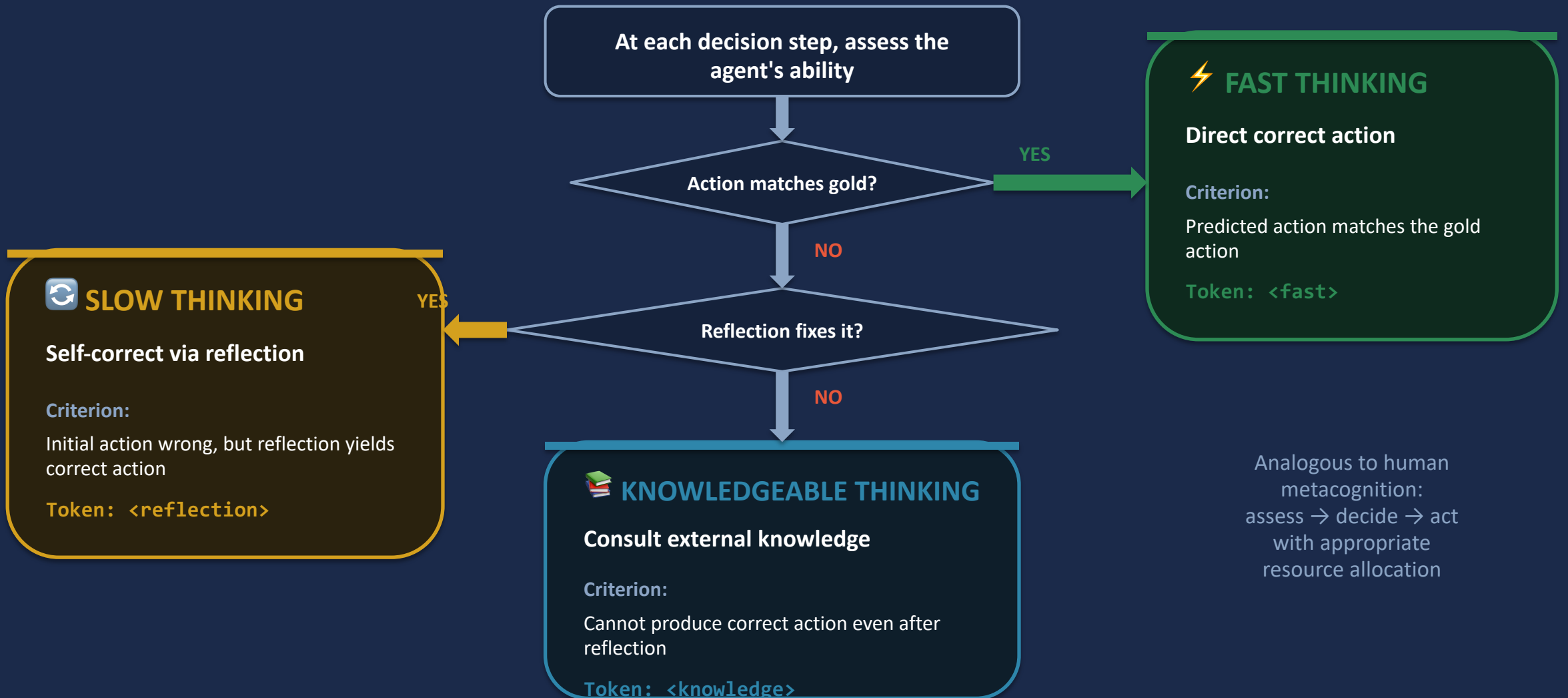
KnowAgent, WKM

Injects knowledge at every step whether needed or not

All three share the same flaw: no situational awareness

Three Situational Modes

Fast Thinking · Slow Thinking · Knowledgeable Thinking



KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

1

Self-Awareness Data Construction

Agent self-explores environments
Heuristic criterion classifies each step:

- Action = gold → <fast>
- Reflection fixes → <reflection>
- Neither → <knowledge>

Special tokens mark situations

2

Two-Stage Training

Stage A: Supervised Fine-Tuning (SFT)
Teaches initial self-awareness patterns

Stage B: Reward-based Policy Opt. (RPO)
Refines token generation quality
Learns when to deploy each mode

3

Self-Aware Inference

Agent generates special tokens at runtime

<fast> → Direct action execution

<reflection> → Trigger self-reflection

<knowledge> → Retrieve from KB

Autonomous resource regulation

Key Insight: Only heuristic labeling on self-explored trajectories — no expensive human annotation or massive external feedback required

Qualitative Examples: KnowSelf vs. Baselines

KnowSelf correctly heats a mug and places a saltshaker where baselines fail

Task: "put a hot mug in cabinet"

KnowSelf (Llama-8B) ✓

Generates <reflection> token after opening microwave.
Realizes it should heat the mug using the open microwave.
→ **Correctly heats mug, then places in cabinet.**

OpenAI-O1 ✗

Opens microwave, then incorrectly tries to take an egg from inside it instead of heating the mug.
No situational awareness — lacks reflection trigger.

Mode: Slow Thinking

Task: "put a saltshaker in drawer"

KnowSelf (Llama-8B) ✓

Generates <knowledge> token for this step.
Retrieves: place object without removing unrelated items.
→ **Correctly places saltshaker in the open drawer.**

DeepSeek-R1 ✗

Reflects that the drawer is open, then incorrectly tries to close it first — which fails.
Reflection without knowledge is insufficient here.

Mode: Knowledgeable Thinking

ALFWorld Results: KnowSelf Achieves 84.33% with Llama-8B

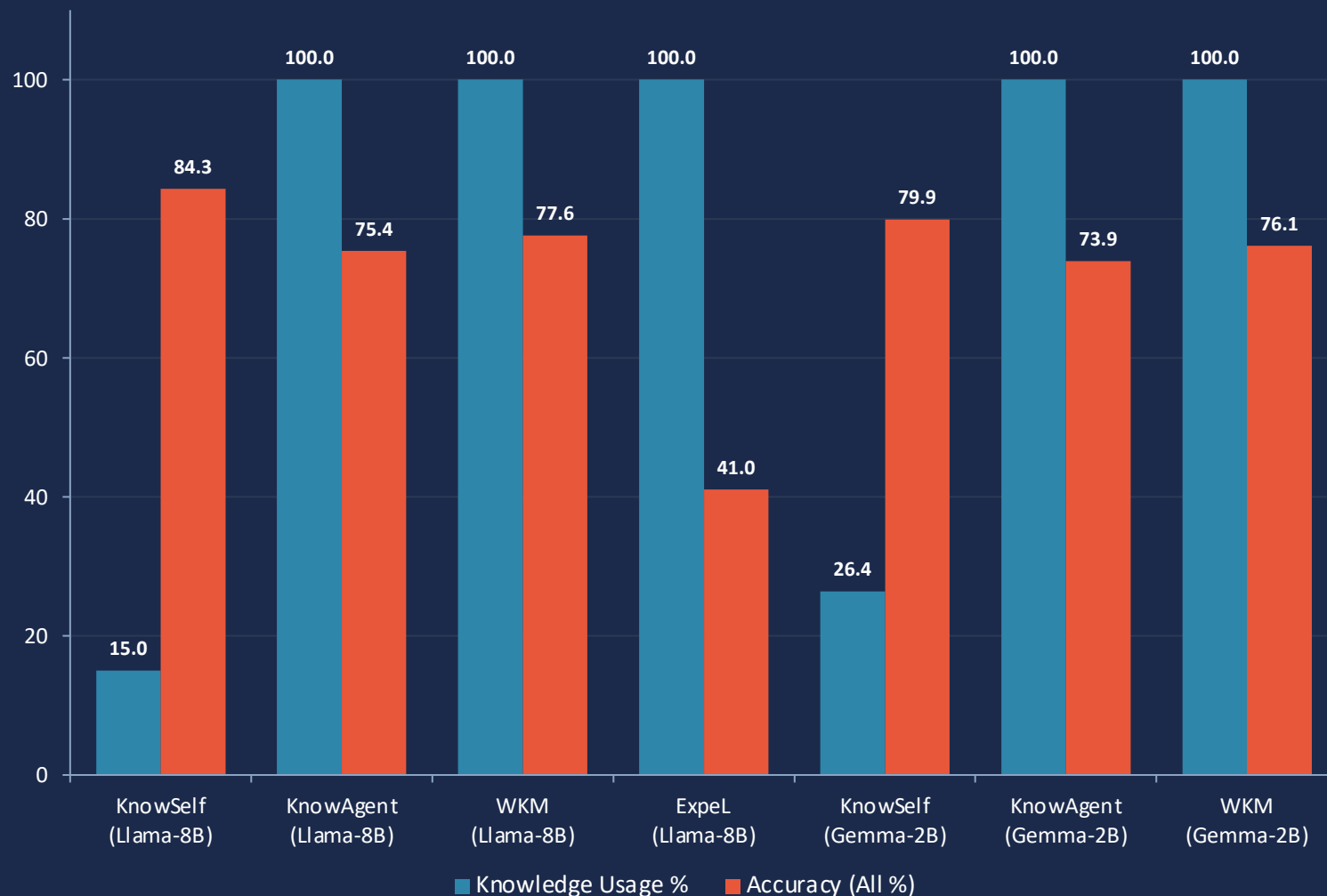
Using only 15% knowledge — outperforming methods that use 100%

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) across six ALFWorld task types | KnowSelf rows highlighted in green | Know% = proportion of steps using external knowledge

Minimal Knowledge, Maximum Impact

KnowSelf's efficiency advantage: ~15–26% knowledge usage vs. 100% for full-knowledge methods



Key Efficiency Insights

Llama-8B KnowSelf:

15% knowledge → 84.33% accuracy

vs. KnowAgent: 100% → 75.37%

vs. WKM: 100% → 77.61%

Gemma-2B KnowSelf:

26% knowledge → 79.85% accuracy

vs. KnowAgent: 100% → 73.88%

vs. WKM: 100% → 76.12%

Scales across model sizes:

Even Gemma-2B with KnowSelf matches GPT-4o ExpeL (79.85%)

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful and suboptimal

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf, confirming more knowledge is not always better.

2

Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories suffices — no expensive human annotation or massive external feedback required.

3

KnowSelf (Llama-8B) outperforms GPT-4o baselines

84.33% vs. GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) — a smaller open model with selective knowledge beats larger models.

4

Only 15–26% knowledge usage needed vs. 100% for competitors

Selective knowledge retrieval dramatically reduces inference cost while improving or maintaining accuracy.

5

The paradigm generalizes across model scales

Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL — the approach scales down effectively.

Future Directions

- Extending to multi-agent and open-domain planning tasks beyond ALFWorld
- Exploring richer metacognitive taxonomies beyond the three-situation framework
- Integrating dynamic knowledge retrieval with retrieval-augmented generation (RAG) pipelines
- Studying transfer of self-awareness capabilities across task domains and model architectures